

Satellite-first water contamination intelligence for a water hackathon

Executive assessment

Your idea is strong, but only if you **reframe it from “detect toxins directly from space” to “detect water anomalies, rank plausible sources, and generate an authority-ready evidence packet.”** With Copernicus, openEO, and Sentinel Hub, a 3-day MVP can reliably detect or infer **water extent changes, turbidity, total suspended matter, chlorophyll-a, trophic state, floating cyanobacteria probability, oil-slick candidates, and nearby industrial sites**. It cannot reliably identify many of the specific substances in your notes—such as arsenic, lead, cyanide, vinyl chloride, calcium, magnesium, lithium, cobalt, uranium/plutonium decay products, Legionella, cholera, or enterobacteria—directly from the satellite imagery alone. Those require in-situ probes, water sampling, or laboratory confirmation. ¹

That makes the winning product concept much clearer: **an early-warning and investigation platform**, not a courtroom-grade accusation engine. The app should help a utility, inspectorate, municipality, NGO, or civil-protection team answer four questions fast: **Where is the anomaly? How severe is it? Which dischargers or land uses are nearby? What evidence should be collected next?** That framing is scientifically defensible, legally safer, and much easier to prototype in a weekend. ²

What the satellites can actually see

The most practical way to locate risky water environments is to combine **water-body detection** with **water-quality anomaly layers**. Copernicus exposes inland-water products, and the current 300 m lake-water-quality product implemented by the Joint Research Centre ³ and the European Environment Agency ⁴ on behalf of the European Commission ⁵ includes bands for **chlorophyll-a concentration, floating cyanobacteria probability, turbidity, total suspended matter, and trophic state**. Sentinel-2 L2A also exposes the spectral bands needed to compute custom water indices over smaller water bodies at finer spatial detail. ⁶

For a hackathon prototype, the most useful “traffic-light” view is this:

Signal	Prototype fit	What it tells you
Water extent / shoreline change	Strong	Flooding, shrinking reservoirs, new plumes, shoreline retreat
Turbidity / suspended matter	Strong	Sediment loads, wastewater plumes, erosion, post-rain contamination
Chlorophyll-a / cyanobacteria risk	Strong	Eutrophication, bloom risk, water-treatment stress

Signal	Prototype fit	What it tells you
Oil slick candidates	Medium	Surface petroleum films, especially with radar and expert review
Atmospheric NO ₂ / SO ₂ / CO / HCHO	Medium for context	Regional industrial-air footprint, not proof of local water discharge
Heavy metals, pathogens, pH, hardness, VOCs, radionuclides	Weak to none	Needs sampling and lab confirmation

This reflects the current CLMS lake-water-quality bands, Sentinel-2 water-quality scripts, Sentinel-1 oil-spill detection use cases, Sentinel-5P atmospheric products, and the well-established limitation that satellites are far better at **optically active water-quality proxies** than at direct retrieval of many dissolved chemicals or pathogens. ⁷

This is the most important scientific correction to your original draft: **you should not search satellite imagery for “chemical elements” in the same way you would search a lab report**. In your stack, the satellite is usually seeing **optical, thermal, or surface-structure consequences** of contamination, not the contaminant itself. It can see greener water, hotter water, murkier water, altered surface reflectance, an oil film, or a change in extent. It usually cannot see “arsenic = yes/no” or “cholera = yes/no” with hackathon-grade confidence. ⁸

The timing also needs honest wording. This is **not real-time** in the emergency-services sense. Under nominal conditions, Sentinel-2 offers a roughly **5-day revisit** with the twin-satellite setup and is limited by clouds; Sentinel-3 OLCI offers **higher temporal frequency** but coarser spatial resolution; Sentinel-5P provides **daily global atmospheric coverage** at multi-kilometer pixels; Copernicus lake-water-quality composites are near-real-time but still arrive on the order of **days after the synthesis period**, not seconds after pollution happens. Sentinel-1 radar is valuable because it is all-weather and day-night, but revisit and data availability still depend on acquisition planning and operational status. The right promise is **near-real-time alerting and periodic surveillance**, not live incident truth. ⁹

Prototype architecture for the hackathon

The best prototype is a **web app**, not a native mobile app. Your core user interaction is a map, layered evidence, time-series comparisons, and facility overlays. A shareable URL is better for judges, mentors, regulators, and NGOs; it is faster to build; and existing hackathon examples in the CASSINI ecosystem already lean heavily toward notebooks plus web dashboards and interactive maps. Mobile should come later for field photos and citizen reports. ¹⁰

A defensible 72-hour architecture looks like this. First, use **water extent** from Copernicus water-body layers or Sentinel-2/Sentinel-1 segmentation to isolate water pixels. Second, pull **water-quality anomaly features** from CLMS Lake Water Quality or from Sentinel-2-derived custom indices. Third, enrich the map with nearby facilities from the European Industrial Emissions Portal ¹¹ and with industrial footprints, land use, wastewater plants, and discharge-adjacent features from OpenStreetMap ¹². Fourth, rank candidate sources with a transparent score that considers distance, upstream position, wind or runoff direction, permit type, and whether the anomaly appeared after rain or after repeated dry weather. Fifth, output a human-readable evidence card instead of a legal accusation. ¹³

A very practical first API example is to request the precomputed Copernicus lake-water-quality product through Sentinel Hub's Process API and render chlorophyll, turbidity, and floating-cyanobacteria probability as a three-band raster:

```
POST https://sh.dataspace.copernicus.eu/process/v1
{
  "input": {
    "bounds": {
      "bbox": [21.38, 41.96, 21.56, 42.09],
      "properties": { "crs": "http://www.opengis.net/def/crs/OGC/1.3/CRS84" }
    },
    "data": [
      {
        "type": "byoc-5c2c9b2c-2893-41d9-b2bc-fbd6e5b8b31d"
      }
    ]
  },
  "output": {
    "width": 512,
    "height": 512,
    "responses": [
      { "identifier": "default", "format": { "type": "image/tiff" } }
    ]
  },
  "evalscript": "//VERSION=3\nfunction setup(){return {input:[{bands: ['CHLAMEAN', 'TMEAN', 'FCBPROB']}],output:{bands:3,sampleType:'FLOAT32'}}}\nfunction evaluatePixel(s){return [s.CHLAMEAN, s.TMEAN, s.FCBPROB]}"
}
```

This structure follows the documented Sentinel Hub Process API request format and the current CDSE BYOC access pattern for the Lake Water Quality 2024-present product. ¹⁴

A second example is to use openEO to load the exact collections you need for a scalable processing pipeline:

```
import openeo

conn = openeo.connect("openeo.dataspace.copernicus.eu").authenticate_oidc()

cube = conn.load_collection(
    "SENTINEL2_L2A",
    spatial_extent={"west": 21.38, "south": 41.96, "east": 21.56, "north": 42.09},
    temporal_extent=["2026-04-01", "2026-04-24"],
    bands=["B03", "B04", "B05", "B08", "SCL"],
)
```

```

    max_cloud_cover=60
)

# Next steps in the process graph:
# - cloud masking from SCL
# - water masking
# - derive water indices or export inputs for anomaly scoring
# - batch over many AOIs

```

openEO currently documents the core collection IDs you need, supports `load_collection` as its main datacube entry point, and includes examples for reusable user-defined processes and larger-scale ML workflows. ¹⁵

Evidence, causation, and legal design

The biggest hole in the original concept is **causation**. A satellite anomaly by itself does **not** prove that a factory caused contamination. Water pollution often comes from agriculture, urban runoff, untreated wastewater, mining, landfill leachate, litter, atmospheric deposition, and storm-driven nonpoint runoff, not only from industrial plants. Rainfall timing, basin topography, river connectivity, season, reservoir operations, and upstream land use can all change the signal you see from space. ¹⁶

A scientifically credible product therefore needs an **evidence chain**. The minimum chain is: a detected anomaly in a water body; a before/after time series showing when it emerged; hydrological logic showing which potential sources are upstream or hydrologically connected; nearby industrial and wastewater facilities with permit context; nonpoint-pressure context such as agriculture or storm runoff; and, before any public allegation, **field verification** with samples or official inspection. If you can add upstream/downstream grab samples or an intake sensor later, your platform becomes much stronger. ¹⁷

Legally, the safe design choice is to avoid public statements like **“Factory X polluted River Y.”** Under EU water, industrial-emissions, environmental-liability, and environmental-crime frameworks, public authorities determine violations, enforce permits, and pursue remediation or prosecution. Your app should therefore use labels such as **“anomaly detected,” “candidate sources nearby,” “field verification required,”** and **“regulatory review recommended.”** That is both more honest and more defensible. ¹⁸

There is also a genuine litigation and platform-risk angle here. As an inference from the fact that anti-defamation regimes exist across Council of Europe states and that online claims can trigger civil proceedings, a startup that publicly accuses a named plant without corroboration could expose itself to legal challenge. The safer product architecture is a **review queue**: machine flags, human analyst reviews, then authority-facing escalation. ¹⁹

For reporting, the app should not auto-email random ministries. It should generate a **jurisdiction-specific evidence packet** containing the coordinates, images, anomaly metrics, nearby regulated sites, weather/runoff context, and confidence score, then route that packet to the appropriate environmental inspectorate, river-basin authority, drinking-water utility, public-health body, or emergency spill channel. Public rights to environmental information and participation are also recognized in the Aarhus framework, so there is room for citizen-facing reporting, but the high-value feature is still the structured evidence pack. ²⁰

Product strategy and scaling

For business scale, the best proposal is **not** a consumer “naming and shaming” app. It is a **B2G or B2B2G risk-intelligence platform** for water utilities, municipalities, basin authorities, environmental inspectorates, ports, aquaculture operators, insurers, ESG teams, and NGOs. Those buyers care about faster triage, better inspection targeting, reservoir protection, reputational risk reduction, and lower sampling costs. That business framing is more credible than “citizens stop pollution from their phones,” especially in a regulated environmental domain. ²¹

On the product side, I would define three user roles. The first is the **analyst/regulator dashboard**, which sees the full evidence chain and export tools. The second is the **utility/operator dashboard**, focused on drinking-water intakes, reservoirs, and bathing-water alerts. The third is a **citizen reporting layer**, but only as supplemental evidence: photo, time, location, smell/color description, optional contact, and moderation. Citizens should help you discover leads, not define ground truth. ²²

You asked whether you should train a model. My honest answer is: **do not make the MVP depend on a custom model that claims to prove industrial guilt**. In a 3-day build, the right approach is an explainable rules-and-threshold engine with optional lightweight anomaly detection. Later, when you have labeled incidents, you can add a supervised ranking model. The literature on heavy-metal retrieval explicitly notes that AI potential is constrained by the scarcity of sufficiently rich concentration/spectral datasets, and official openEO examples show that ML is useful once preprocessing, masking, and features are already solid. ²³

That is also why you should not “pretend it is super advanced AI.” The smarter pitch is: **transparent environmental intelligence**. Judges in this domain usually reward credibility over inflated claims, and the existing CASSINI examples already show that notebook-backed analytics, web maps, and explainable scoring are acceptable hackathon patterns. ²⁴

Contaminants and remediation reality

Your contaminant list mixes four very different categories: **optically active water-quality signals, microbial/pathogen risks, dissolved toxic chemicals/metals, and nuisance chemistry such as hardness and pH**. Those categories must not be handled as if they were equivalent. The World Health Organization ²⁵ treats fecal contamination and unsafe water as a major public-health issue, while the Centers for Disease Control and Prevention ²⁶ notes that *Legionella* is naturally present in freshwater but becomes a major disease risk mainly in **building water systems**, where temperature, biofilm, sediment, water age, and disinfectant residual matter. That means *Legionella* and cholera belong in your platform as **risk contexts**, not direct satellite detections. ²⁷

For nutrients and bloom-related problems, your platform is on much firmer ground. Satellite-derived chlorophyll, cyanobacteria probability, turbidity, and trophic state are directly aligned with eutrophication and harmful algal bloom surveillance. Existing public systems such as EPA's CyAN use satellites as **early-warning indicators**, and Sentinel Hub custom scripts already expose practical workflows for chlorophyll-a, cyanotoxins-related indicators, turbidity, cDOM, and TSS over inland waters. ²⁸

For metals and highly toxic chemicals, the science is less forgiving. Arsenic is associated with cancer, skin lesions, cardiovascular disease, diabetes, and developmental harms, and the WHO emphasizes that the most important action is to **prevent further exposure by providing safe water**. Lead remains unsafe even at very low exposure, especially for children. Cyanide is highly acutely toxic and becomes a serious drinking-water issue mainly in **spill or industrial-release scenarios**. These are high-consequence contaminants, but in your satellite stack they are mostly **investigation targets for sampling**, not direct retrieval products. Treatment can involve source isolation, adsorption or coagulation, ion exchange, precipitation, membranes, and other engineered processes depending on chemistry and concentration. ²⁹

For hardness, calcium, magnesium, CO₂-related acidity, and pH, the right answer is even more important because it affects your science narrative. **There is no serious remediation rule like “the water has calcium, so add anti-calcium chemical.”** Hardness is mostly caused by calcium and magnesium salts; it is often an operational and infrastructure issue rather than an acute toxic emergency; and treatment is usually done with softening processes such as cation exchange or other standard water-treatment methods. pH is a master variable that changes toxicity and biology, but it is not something your satellite stack can estimate robustly enough for compliance without local calibration. ³⁰

For petroleum derivatives, oil films, VOCs such as vinyl chloride, pesticides, radionuclides, and battery-related metals such as lithium and cobalt, your platform should again separate **detection** from **confirmation**. Sentinel-1 can be useful for surface oil-spill candidates, but wastewater chemistry, vinyl chloride, pesticides, uranium, lithium, and cobalt typically need on-site or laboratory measurement. Petrol- and solvent-contaminated sites are also classic cases where cleanup design depends on the intended land and water use, so the correct product move is to flag the anomaly, identify candidate sources, and recommend the next sampling or containment step—not to prescribe a universal remedy from the map. ³¹

Recommended pitch and demo

If I were turning your concept into a submission for CASSINI Hackathons ³², I would pitch it like this: **a web platform that detects water-quality anomalies from Copernicus data, correlates them with nearby regulated facilities and land-use pressures, and produces evidence packets for utilities and authorities before contamination becomes a public-health crisis**. The core promise is not “we identify every toxin from space.” The core promise is **“we cut the time from anomaly to investigation.”** ³³

Your demo should show exactly one strong workflow from end to end. The ideal flow is: search a lake or reservoir, display water extent and anomaly layers, click a hotspot, inspect chlorophyll/turbidity/cyanobacteria metrics and a short time trend, show nearby industrial or wastewater facilities, display a confidence score and “possible drivers” panel, then export a concise evidence card. That is enough for a convincing MVP. You do not need full automation, and you do not need to prove legal liability. ³⁴

Among the side ideas you mentioned, the most compatible add-on is **drinking-water intake protection** or **bathing-water / algal-bloom risk**, because those use the same anomaly stack and create an immediate public-health story. Your wildfire idea is viable as a separate challenge, but I would not mix forest-fire detection into the core pitch unless you frame it as **post-fire runoff and sediment risk to reservoirs**, which is still a water story. For the main submission, keep the message tight: **protect source waters, rank risks, support action**. ³⁵

Open questions and limitations

The remaining unknowns are practical rather than conceptual.

- You still need to choose a **pilot geography** and one or two water-body types, because small rivers, reservoirs, and large lakes do not behave the same way in EO products.
- You need a **jurisdiction routing table** for reporting, because authority chains differ by country and incident type.
- You need at least some **ground-truth reference data** if you want to calibrate local thresholds or train later ML models.
- You should confirm API quotas, processing costs, and the exact datasets you can call from your keys during the weekend.
- Most importantly, you should decide whether your judges will respond better to **environmental surveillance**, **public-health protection**, or **regulatory decision support**. The product can serve all three later, but the pitch should choose one.

1 3 6 7 34 https://documentation.dataspace.copernicus.eu/APIs/SentinelHub/Data/clms/bio-geophysical-parameters/water-bodies/lake-water-quality/lwq-nrt_global_300m_10daily_v2.html
https://documentation.dataspace.copernicus.eu/APIs/SentinelHub/Data/clms/bio-geophysical-parameters/water-bodies/lake-water-quality/lwq-nrt_global_300m_10daily_v2.html

2 https://environment.ec.europa.eu/law-and-governance/environmental-compliance-assurance_en
https://environment.ec.europa.eu/law-and-governance/environmental-compliance-assurance_en

4 16 <https://www.epa.gov/nps/basic-information-about-nonpoint-source-nps-pollution>
<https://www.epa.gov/nps/basic-information-about-nonpoint-source-nps-pollution>

5 31 <https://sentinels.copernicus.eu/web/success-stories/-/copernicus-sentinel-1-data-enable-oil-spill-detection-in-south-eastern-mediterranean-sea>
<https://sentinels.copernicus.eu/web/success-stories/-/copernicus-sentinel-1-data-enable-oil-spill-detection-in-south-eastern-mediterranean-sea>

8 11 <https://www.sciencedirect.com/science/article/abs/pii/S0141113622001465>
<https://www.sciencedirect.com/science/article/abs/pii/S0141113622001465>

9 https://custom-scripts.sentinel-hub.com/custom-scripts/sentinel-2/mago_water_quality_monitoring_tool/
https://custom-scripts.sentinel-hub.com/custom-scripts/sentinel-2/mago_water_quality_monitoring_tool/

10 https://github.com/JWLZGG/Race_Publica
https://github.com/JWLZGG/Race_Publica

12 17 <https://industry.eea.europa.eu/industrial-emissions/dataset>
<https://industry.eea.europa.eu/industrial-emissions/dataset>

13 25 https://documentation.dataspace.copernicus.eu/APIs/SentinelHub/Data/clms/bio-geophysical-parameters/water-bodies/water-bodies/wb_global_300m_monthly_v2.html
https://documentation.dataspace.copernicus.eu/APIs/SentinelHub/Data/clms/bio-geophysical-parameters/water-bodies/water-bodies/wb_global_300m_monthly_v2.html

- 14 <https://documentation.dataspace.copernicus.eu/APIs/SentinelHub/Process.html>
<https://documentation.dataspace.copernicus.eu/APIs/SentinelHub/Process.html>
- 15 <https://documentation.dataspace.copernicus.eu/notebook-samples/openeo/basics.html>
<https://documentation.dataspace.copernicus.eu/notebook-samples/openeo/basics.html>
- 18 https://environment.ec.europa.eu/topics/water/water-framework-directive_en
https://environment.ec.europa.eu/topics/water/water-framework-directive_en
- 19 <https://rm.coe.int/liability-and-jurisdictional-issues-in-online-defamation-cases-en/168097d9c3>
<https://rm.coe.int/liability-and-jurisdictional-issues-in-online-defamation-cases-en/168097d9c3>
- 20 22 32 https://environment.ec.europa.eu/law-and-governance/aarhus_en
https://environment.ec.europa.eu/law-and-governance/aarhus_en
- 21 <https://industry.eea.europa.eu/>
<https://industry.eea.europa.eu/>
- 23 <https://www.mdpi.com/2072-4292/16/20/3888>
<https://www.mdpi.com/2072-4292/16/20/3888>
- 24 26 <https://github.com/cassinihackathons/intdev-humaid-examples>
<https://github.com/cassinihackathons/intdev-humaid-examples>
- 27 <https://www.who.int/news-room/fact-sheets/detail/drinking-water>
<https://www.who.int/news-room/fact-sheets/detail/drinking-water>
- 28 35 <https://www.epa.gov/water-research/cyanobacteria-assessment-network-cyan>
<https://www.epa.gov/water-research/cyanobacteria-assessment-network-cyan>
- 29 <https://www.who.int/news-room/fact-sheets/detail/arsenic>
<https://www.who.int/news-room/fact-sheets/detail/arsenic>
- 30 <https://www.usgs.gov/water-science-school/science/hardness-water>
<https://www.usgs.gov/water-science-school/science/hardness-water>
- 33 <https://www.cassini.eu/hackathons/tools>
<https://www.cassini.eu/hackathons/tools>